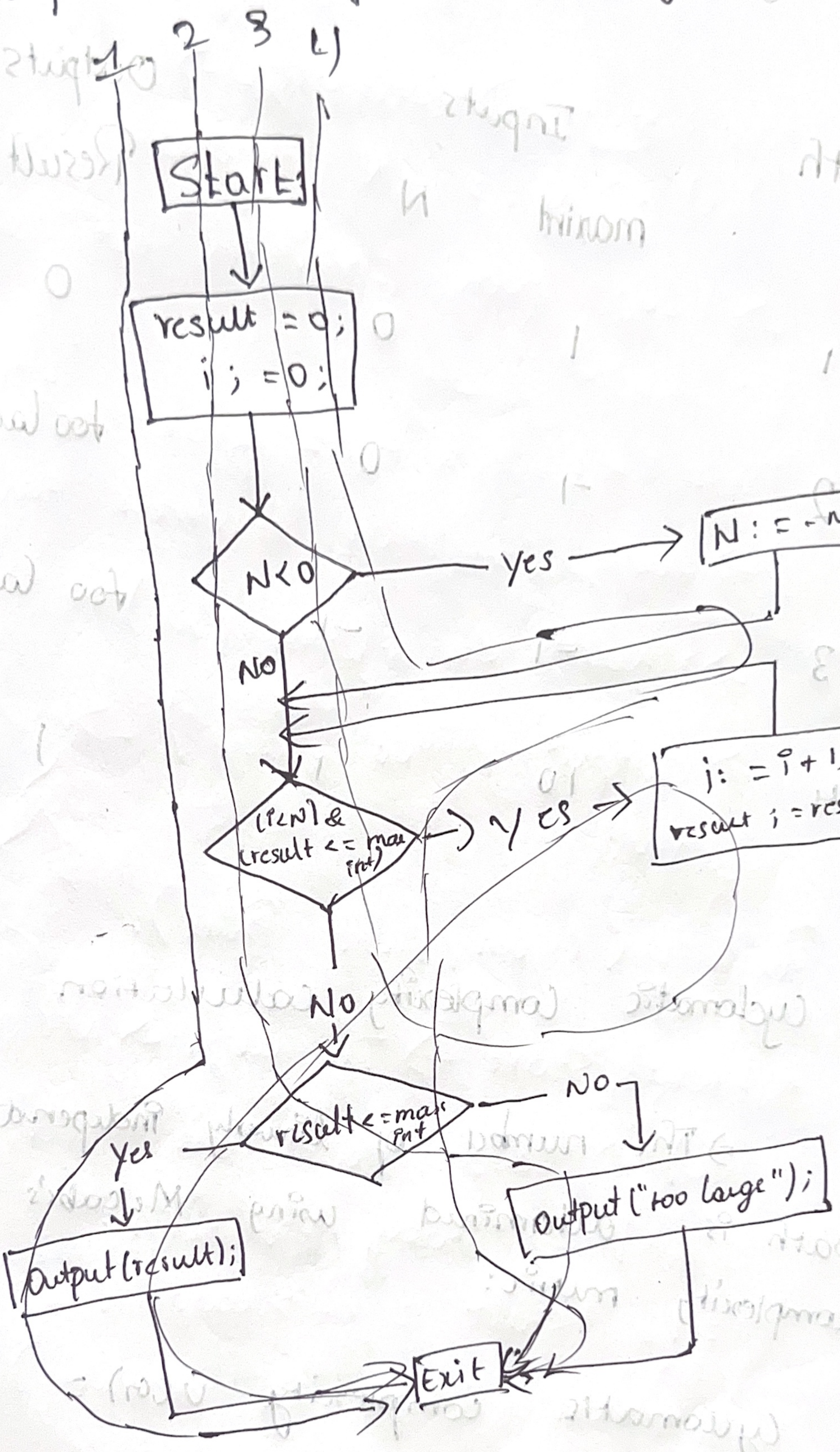
Independent path coverage:



Number of independent paths

= cyclomatic complexity

= number of edges - number of nodes + 2

= 12 - 10 + 2

= 4

Test cases:

Path	Inputs		Outputs
	maxint	N	Result
1	1	0	0
2	-1	0	too large
3	-7	-1	too large
4	10	1	1

1) Cyclomatic Complexity Calculation

⇒ The number of linearly independent path is determined using McCabe's cyclomatic complexity metric:

Cyclomatic complexity V(G) =

* Edges (E): 12 (the directed flow lines connecting the nodes)

* Nodes (N): 10 (the individual process and decision boxes, including start

* Calculation:

12 - 10 + 2 = 4

2) For the 4 Independent Execution paths:

Path 1 (Dark Blue):

* Flow: start → Initialize (result = 0, i = 0) → N < 0 (no) → Loop condition (i < N) and (result <= maxint) (False, since 0 < 0 is false) → result <= maxint (yes, 0 <= 1) → output (result) → Exit.

* Test inputs:

maxint = 1 ; N = 0

* output:

result = 0

Path 2 (Yellow):

*Flow: start → initialize → N < 0 (NO) → Loop condition (False, 0 < 0 is false) → result <= maxint (NO, 0 <= -1 is false) → output ("too large") → Exit.

Path 3 (Black / Gray):

Flow: Start → Initialize → N<0
(yes) → N := -N (negates N to +ve)
→ Loop body Executes at least one
(
i := i+1 ; result := result + i
) → Loop Condition becomes false
→ result <= maxint (NO)
→ output ("too large") →
Exit.

Path 4 (Cyan / Light Blue):-

* Flow: Start → Initialize → N<0
(NO) → Loop condition (true,
0<11 → Body Executes.
i=1, result = 1 → Loop
condition (false, 1<11 is false)
→ result <= maxint (yes,
1<=10) → output (result) →
Exit.